B					
C					
D					
E					
F					

Finals (After Mids):-

Context-Free Grammer Examples:-

- CFG for language that contain words of one length character.

RE = a+b

CFG: $S \rightarrow a|b$

- That contains of two length.

R.E = (a+b)(a+b)

CFG: aa|bb|ab|ba

OR

$S \rightarrow AA$

$A \rightarrow a|b$

$a^+/a^{+}+a$

. At most 3 characters.

R.E = $(\epsilon+a+b)(\epsilon+a+b)(\epsilon+a+b)$

CFG = $S \rightarrow AAA$

$A \rightarrow a|b|\epsilon$

. That contain at least one 'a'. $\Sigma = \{a\}$

R.E = a^+

CFG:

. At least two char. with start & end with same Symbol.

R.E = $(a(a+b)^*a)+(b(a+b)^*b)$

CFG:- $aAa \mid bAb \longrightarrow S$

$A \rightarrow aA \mid bA \mid a \mid b$

. as third last char b;

R.E = $(a+b)^* b (a+b)(a+b)$

CFG:- $S \rightarrow AbBB$

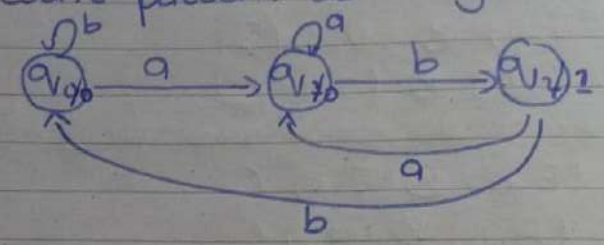
$A \rightarrow aA \mid bA \mid a \mid b$

$B \rightarrow a \mid b$

Transdures OF DFA:-

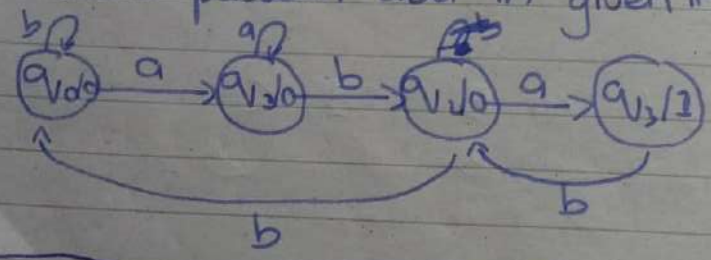
Here we can count the string.

Q. count pattern ab in given input;



Input
state
output

Q: count pattern aba in given input;



CFC TO CFG:-

(i) $a^n b^n$, $n \geq 0 \longrightarrow S \rightarrow aSb \mid \epsilon$

$n \geq 1 \longrightarrow S \rightarrow aSb \mid ab$

(ii) $a^n b^{n+2}$, $n \geq 0 \longrightarrow S \rightarrow aSb \mid bb$

, $n \geq 1 \rightarrow S \rightarrow aSb \mid abbb$

(iii) $a^{2n} b^n$, $n \geq 0 \longrightarrow S \rightarrow aaSb \mid \epsilon$

, $n \geq 1 \longrightarrow S \rightarrow aaSb \mid aab$

put;

(iv) $a^{2n+3} b^n$, $n \geq 0 \rightarrow S \rightarrow aaSb \mid aaa$

(v) $a^m b^n$, $m > n$ $\rightarrow$ $S \rightarrow AS_2$
$m, n \geq 0$ $S_2 \rightarrow aSb \mid \epsilon$
$A \rightarrow aA$

(vi) $\{w \mid n_a(w) = n_b(w)$

$S \rightarrow aSb \mid bSa \mid \epsilon$
$aSb \mid bSa \mid SS \mid \epsilon$

(vii) $\underset{\uparrow \text{Even}}{ww^R} \cup \underset{\uparrow \text{odd}}{w(a+b)w^R}$
$S \rightarrow aSa \mid bSb \mid \epsilon \rightarrow$ Even
$S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon \rightarrow$ odd

(viii) $a^m b^m c^n$, $m, n \geq 0$
$S \rightarrow S_1 C$
$S_1 \rightarrow aS_1 b \mid \epsilon$
$C \rightarrow cC \mid \epsilon$

(ix) $a^m b^n c^x d^n e^m$ $m \geq 0$, $n \geq 0$, $x \geq 0$
$S \rightarrow A$
$A \rightarrow aAe \mid B$
$B \rightarrow bBd \mid bCd$
$C \rightarrow cC \mid \epsilon$

t;

Chomsky Normal Form:-

CFG TO CNF:-

(i) $S \rightarrow aXbX$
$X \rightarrow aY/bY/\epsilon$
$Y \rightarrow X/c$

$S \rightarrow aXbX$
$X \rightarrow aY$
$X \rightarrow bY$
$X \rightarrow \epsilon$
$Y \rightarrow X$
$Y \rightarrow c$

Eliminating ϵ:-

$S \rightarrow aXbX \mid aXb \mid abX \mid ab$
$X \rightarrow aY \mid bY \mid a \mid b$
$Y \rightarrow X \mid c$

Push-Down Automata:-

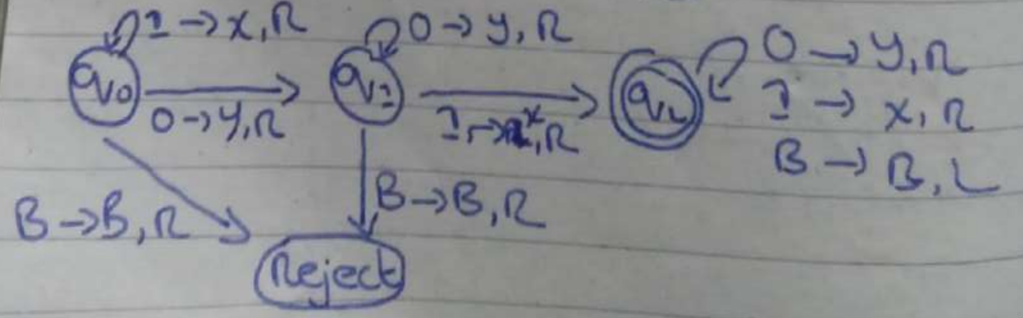
Combination of Finite Machine & Stack.

Introduction To Turing Machines:-

Q: Design a TM over [0,1] that accept "01"

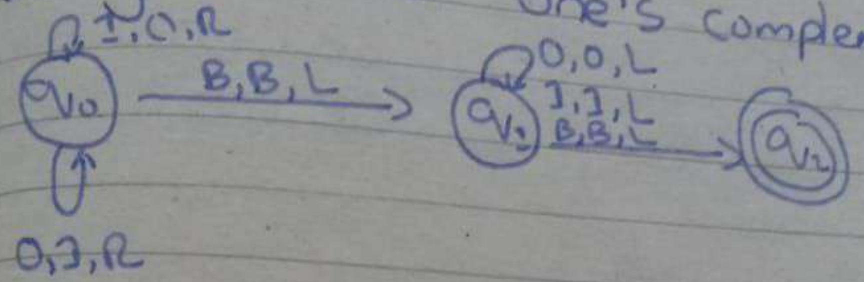
$$(0+1)^* \, 01 \, (0+1)^*$$

B	1	1	0	0	1	0	1	B

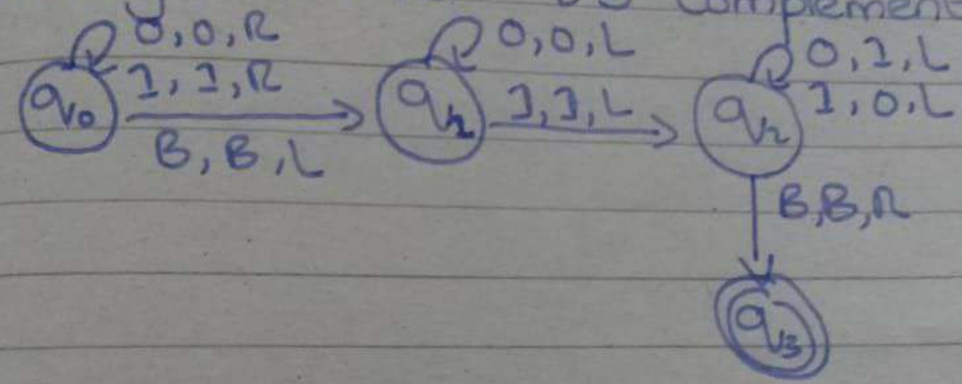


1000101110000
0111010100000

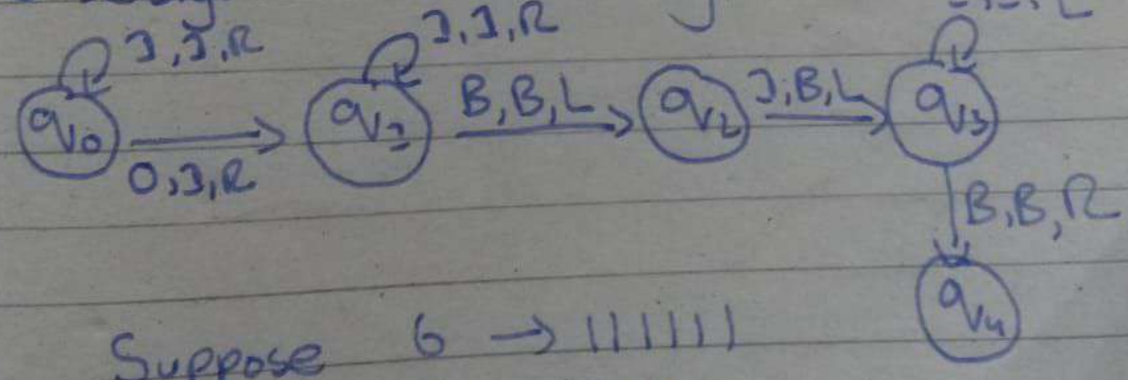
Q: Design TM For One's complement



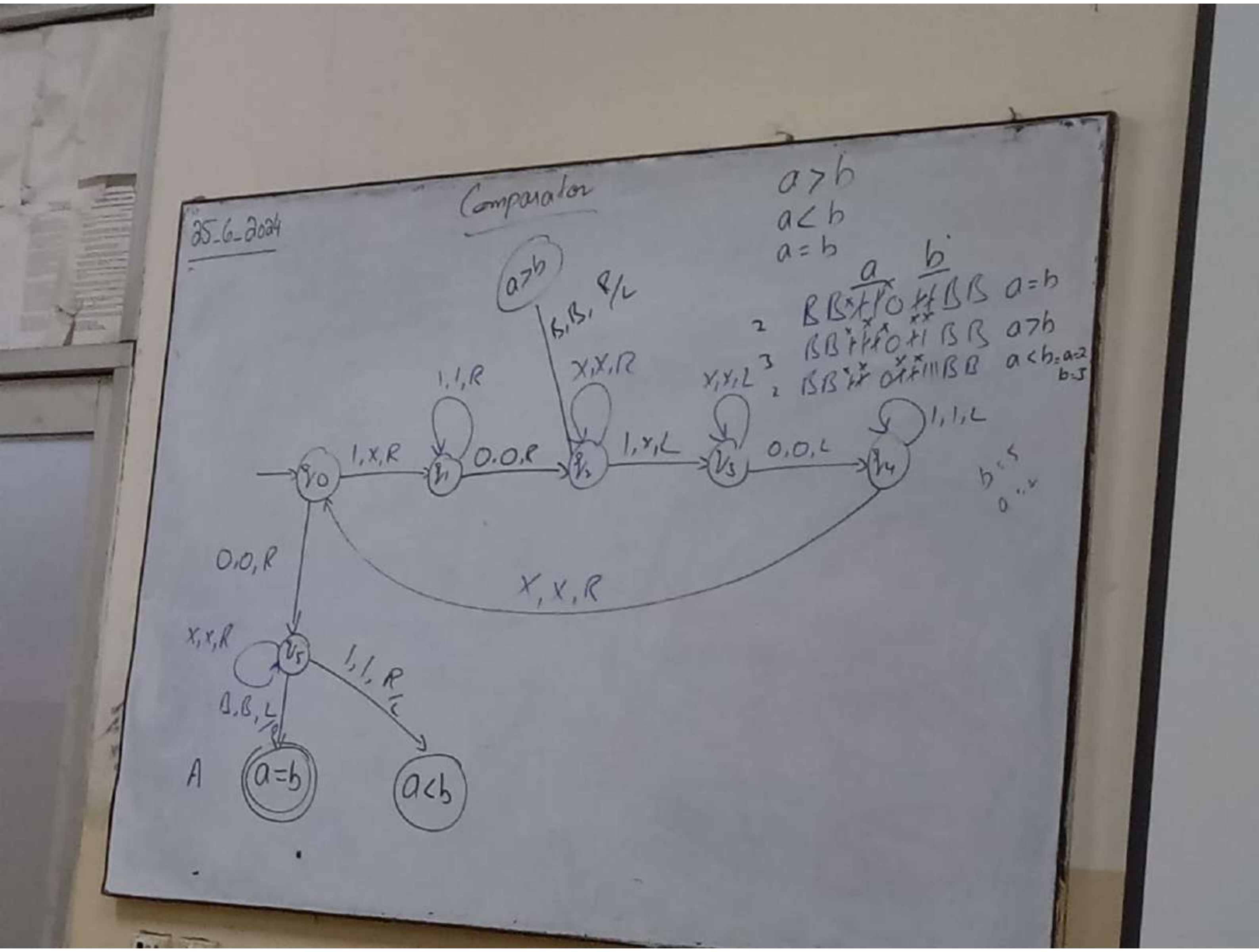
Q: Design TM For 2's complement



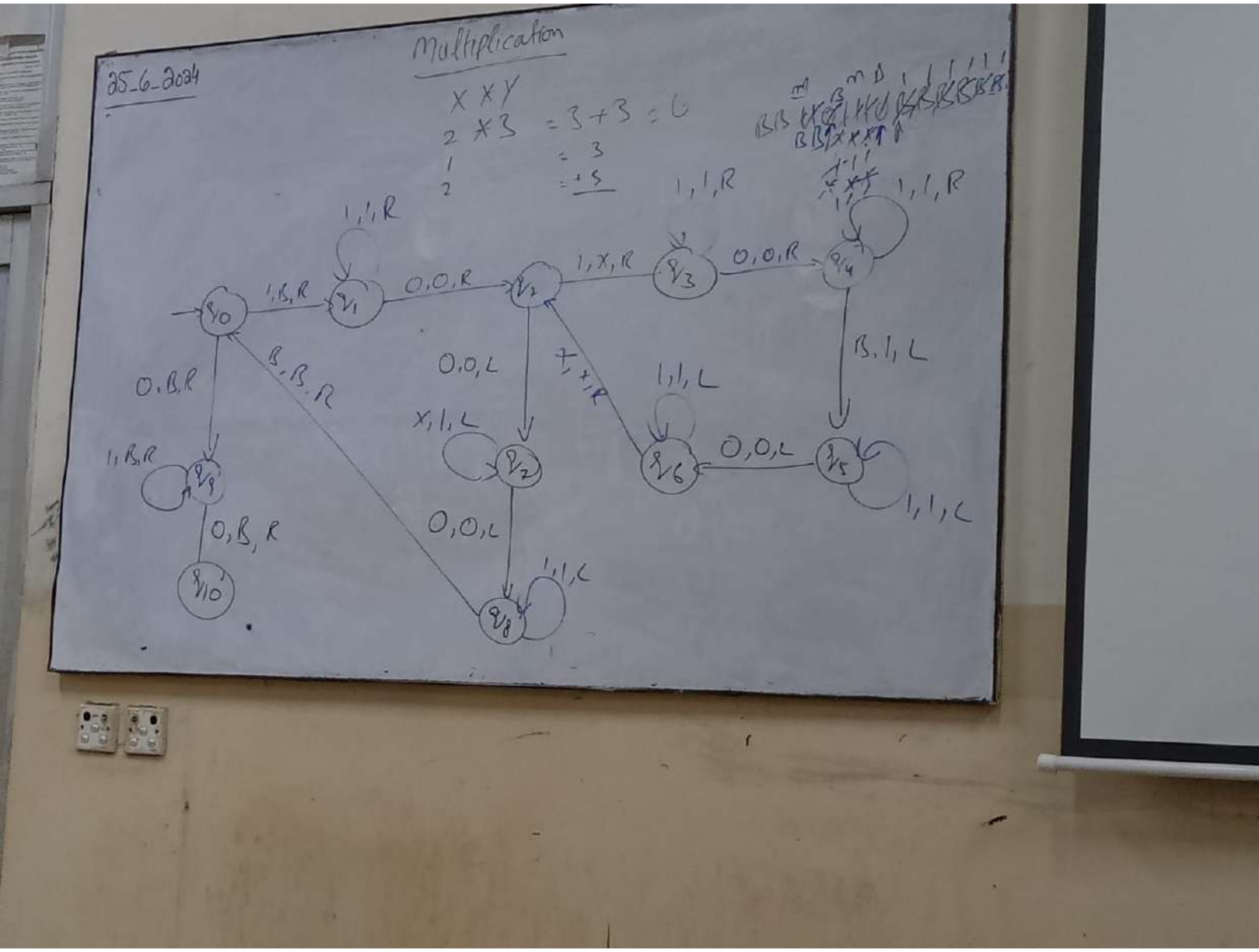
Q: Design TM For Adding



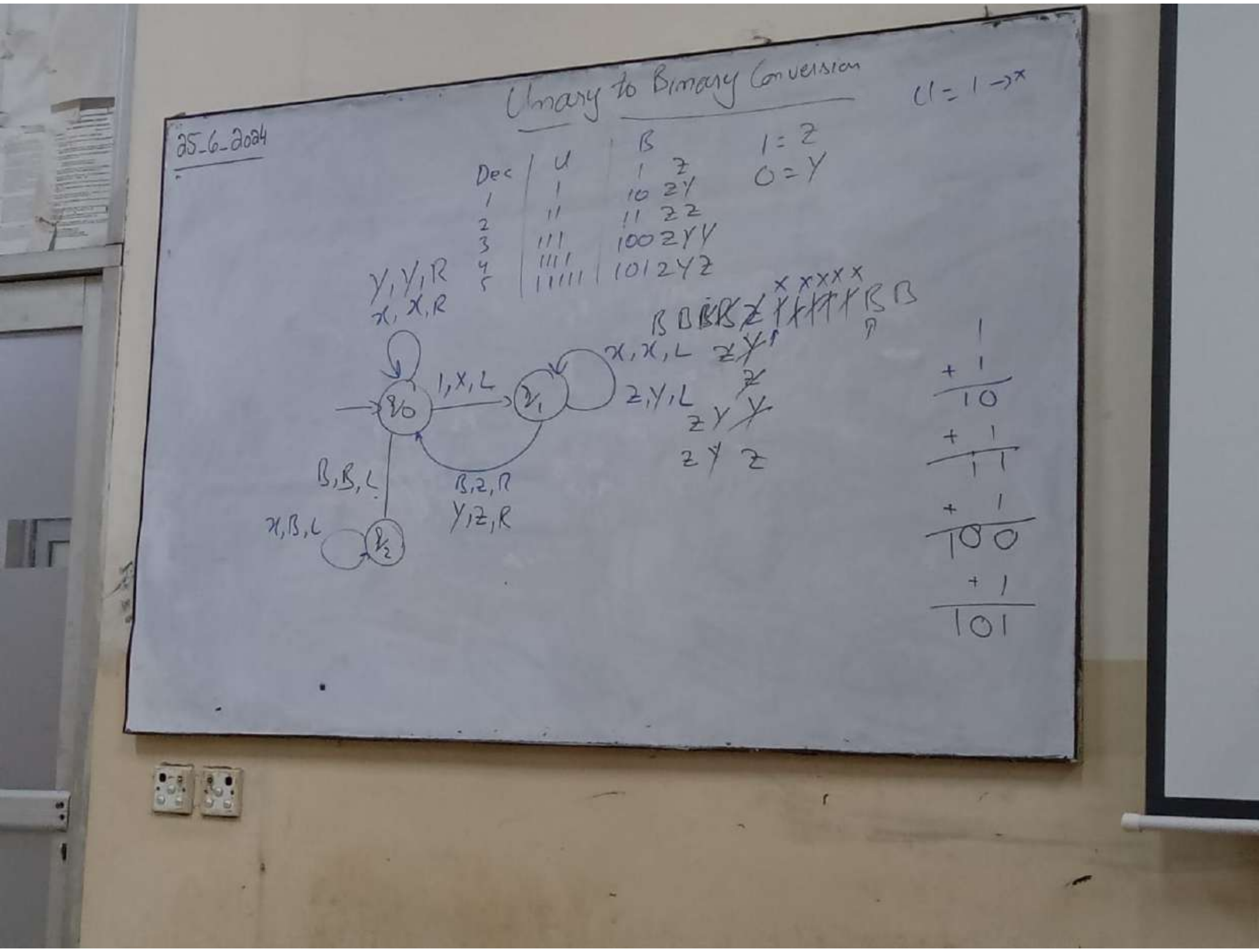
Suppose 6 → 111111
+2 → 11
8

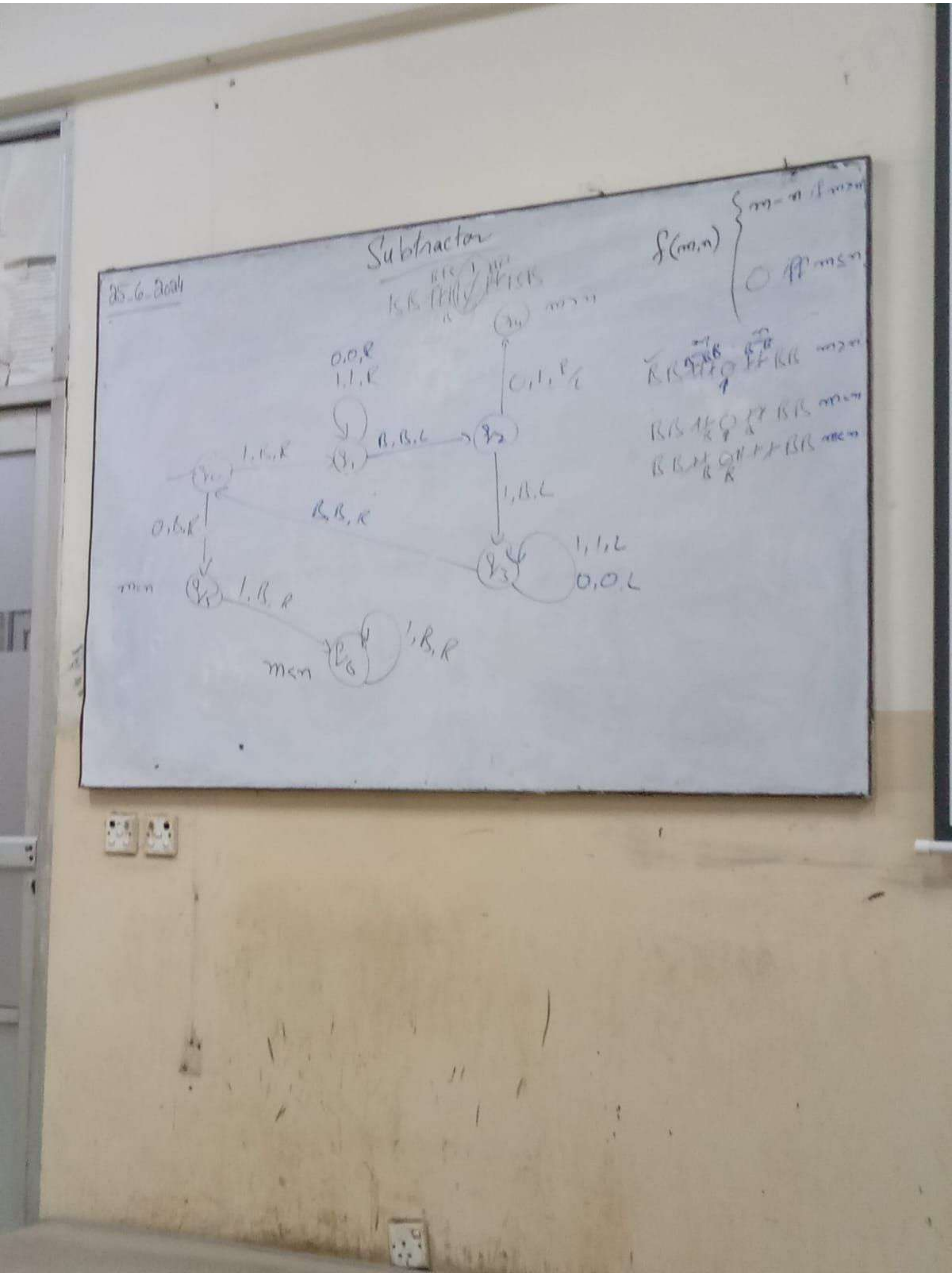


25-6-2024
Comparator
a > b
a < b
a = b
a
b
BBXXXOXXBB a=b
BBXXXOXXBB a>b
BBXXOXXXBB a<b
1,1,R
X,X,R
X,X,L
1,1,L
1,X,R
0,0,R
1,X,L
0,0,L
X,X,R
B,B,R/L
a>b
0,0,R
X,X,R
B,B,L
1,1,R
A
a=b
a<b

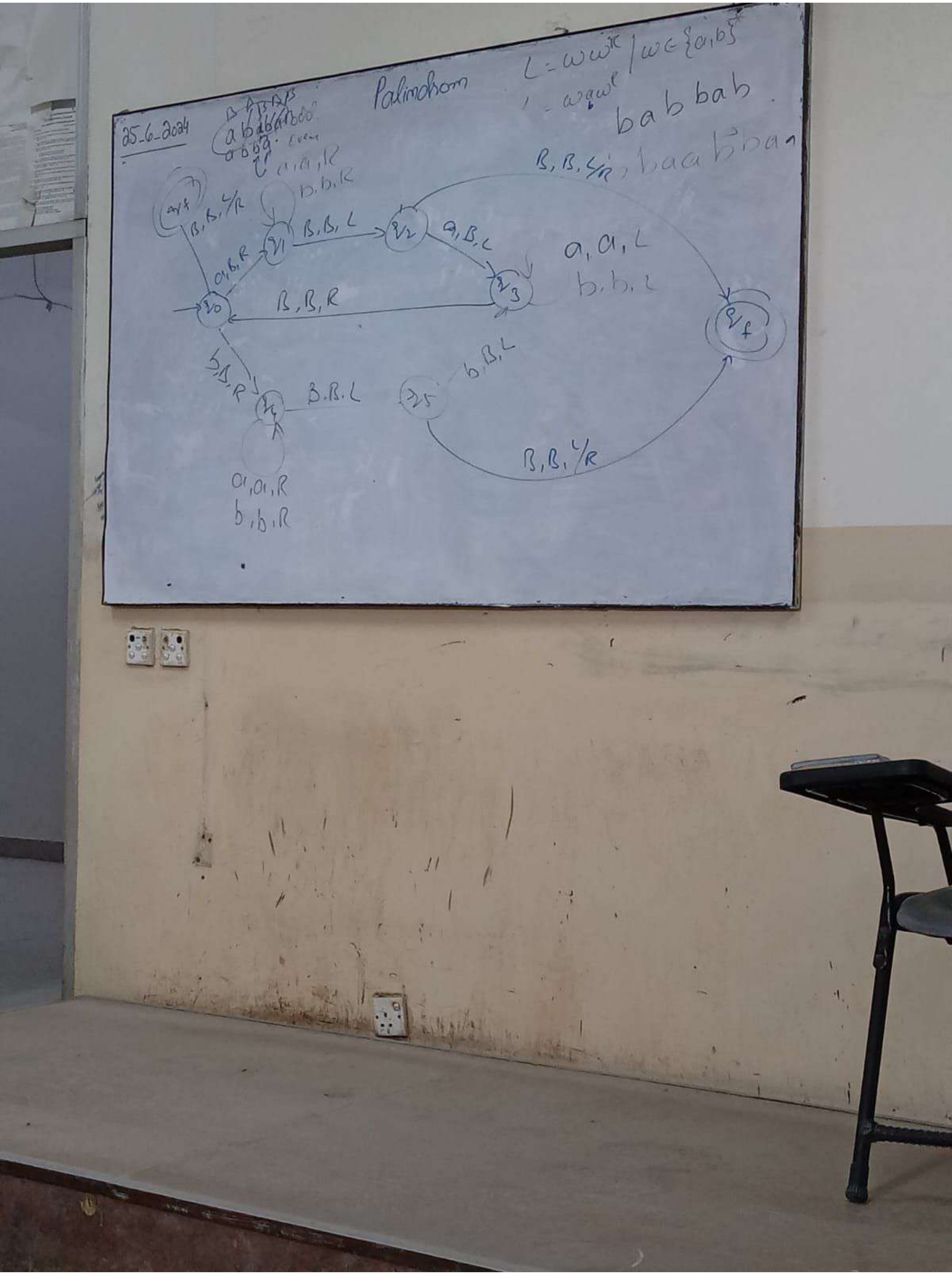


25-6-2024
Multiplication
X * Y
2 * 3 = 3 + 3 = 6
1,1,R
1,B,R
0,0,R
1,X,R
0,0,R
1,1,R
B,1,L
0,0,L
1,1,L
1,1,L
0,0,L
X,1,L
0,0,L
1,1,L
B,B,R
0,B,R
1,B,R
0,B,R


Unary to Binary Conversion
U = 1 → x
25_6_2024
Dec | U | B
1 | 1 | 1 Z
2 | 11 | 10 ZY
3 | 111 | 11 ZZ
4 | 1111 | 100 ZYY
5 | 11111 | 101 ZYZ
1 = Z
0 = Y
Y, Y, R
x, X, R
1, X, L
x, x, L
Z, Y, L
B, Z, R
Y, Z, R
B, B, L
x, B, L
1
+ 1
10
+ 1
11
+ 1
100
+ 1
101



Subtractor
25-6-2024
f(m,n)
0,0,R
1,1,R
1,B,R
B,B,L
1,B,L
B,B,R
0,B,R
1,1,L
0,0,L
1,B,R
m<n



25-6-2024
Palindrom
L = ww^R | w ∈ {a,b}*
bab bab
B,B,L/R
a,a,R
b,b,R
B,B,L
a,B,L
a,a,L
b,b,L
B,B,R
a,B,R
b,B,R
B.B.L
b,B,L
B,B,L/R
a,a,R
b,b,R
q0
q1
q2
q3
q4
q5
qf